

HIMANSHU
Pre-Final Year Undergraduate
Computer Science Engineering
National Institute Of Technology, Kurukshetra

+91-8572849658
✉ himanshu4170@gmail.com
📞 himanshu4170
🌐 himanshu

EDUCATION

- **National Institute of Technology, Kurukshetra** 2027
Bachelor of Technology CGPA: 7.675
- **Yaduvanshi Shiksha Niketan, Narnaul** 2022
Central Board of Secondary Education (XII) Percentage: 89.6
- **Yaduvanshi Shiksha Niketan, Narnaul** 2020
Central Board of Secondary Education (X) Percentage: 89.6

PERSONAL PROJECTS

- **AI-Powered Door Locking System** 2024
A smart door security system using facial recognition with deep learning and embedded simulation
 - Designed and implemented a secure door locking system that uses a **CNN-based facial recognition** model to authenticate authorized users.
 - Trained the model using a custom dataset with **NumPy**, **OpenCV**, and **TensorFlow/Keras**, leveraging data augmentation and preprocessing techniques for robust performance.
 - Simulated hardware logic in **Proteus Design Suite**, including **servo motor control**, **IR sensors**, and **camera integration** for real-time access management.
 - Built the inference pipeline for real-time face detection and classification, optimizing model inference speed using **cv2.dnn** and **Haar cascades** as fallback .
 - Applied concepts from **Computer Vision**, **Embedded Systems**, and **IoT** to design a fully functional prototype.
 - Ensured system reliability under varied lighting conditions and angles, achieving over **95% accuracy** in real-time authentication scenarios.
 - Demonstrated modular system design and scalability for future deployment on microcontroller platforms like **Raspberry Pi** or **ESP32**.
- **The Bank Application** 2024
A desktop-based banking system built using JavaFX and SQLite with an MVC architecture
 - Designed and implemented a secure and user-friendly desktop banking application with features like account creation, authentication, deposits, withdrawals, and transaction history.
 - Built the GUI using **JavaFX** with enhanced visuals using **FontAwesomeFX** icons, ensuring an intuitive and modern user experience.
 - Followed the **Model-View-Controller (MVC)** design pattern for code modularity, scalability, and separation of concerns.
 - Integrated an embedded relational database using **SQLite** with **JDBC** for storing user credentials, account balances, and transaction logs.
 - Used **FXML** for interface design and **JavaFX Controllers** to handle UI logic. Implemented proper input validation, exception handling, and secure login mechanisms.
 - Tools & technologies used: **JavaFX**, **FXML**, **SQLite**, **JDBC**, **FontAwesomeFX**, **OOP**, **MVC Architecture**.

TECHNICAL SKILLS

Languages: C, C++, Java, Python , JavaScript
Developer Tools: VS Code, Git, GitHub, Arduino IDE, Tinkercad, Proteus
Cloud/Databases: SQL
Soft Skills: Communication, Teamwork, Problem-solving
Coursework: Data Structures and Algorithms, Python, Object-Oriented Programming, Digital System Design, Computer Organization and Architecture, Internet of Things, Operating Systems

ACHIEVEMENTS

- **Qualified NDA written exam** National Defence Acadmy(NDA) 2020–2021
- **Solved 300+ questions on Leetcode** Competitive Programming 2024